# BTC 5-Minute Strategy Research Report

Candidate strategies benchmarked against the existing **Sway model**. Generated 2026-06-28 04:43:11 UTC.

## Executive Summary

Trained on **600** historical markets, tested out-of-sample on **250** more-recent markets (test base rate 53.2% UP). Best by accuracy: **MarketPrice** (89.0%). The **Sway baseline is the weakest model** (77.4% accuracy) — worse than simply trusting the live market price (89.0%).

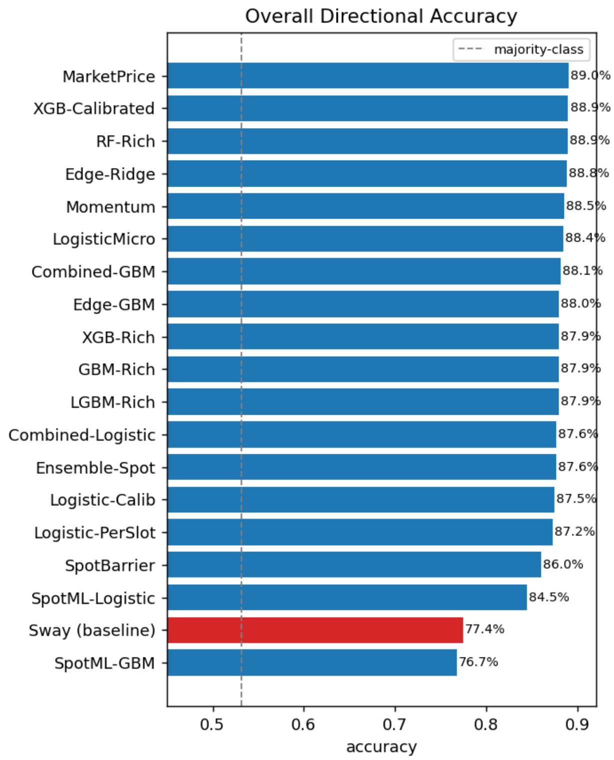
**Headline result.** The biggest edge comes from a signal the prediction-market models never use: the **underlying BTC spot price** (Binance 1s data). A purely analytic first-passage 'barrier' probability computed from the live spot lead and realised volatility is the strongest trader. Crucially, it is re-tested on a second independent window — where most strategies' trading ROI flips sign (noise). The strategies profitable on **both** windows are: Ensemble-Spot, SpotBarrier, Combined-Logistic, Combined-GBM, LogisticMicro. Most robust (best worst-case ROI): **Ensemble-Spot** (+20.5% test / +24.0% val). Statistical accuracy ranking is stable throughout: the Sway baseline is consistently last.

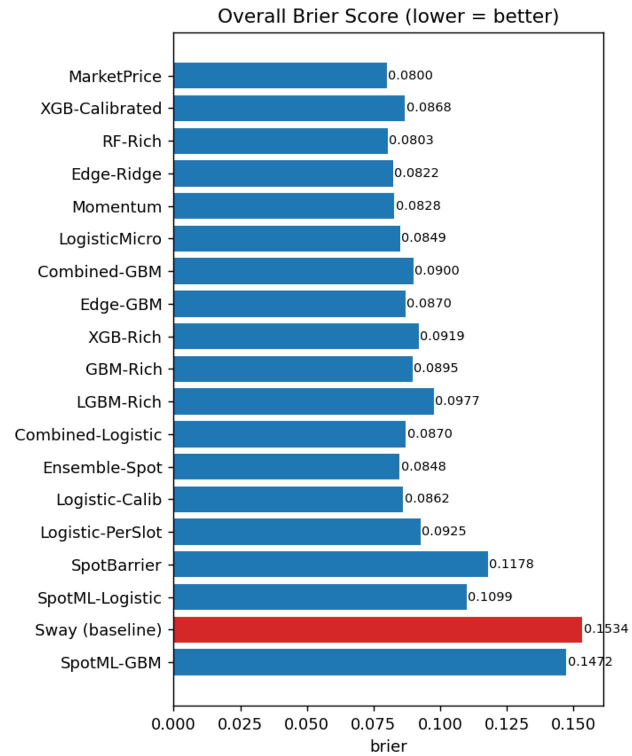
## Overall metrics (out-of-sample)

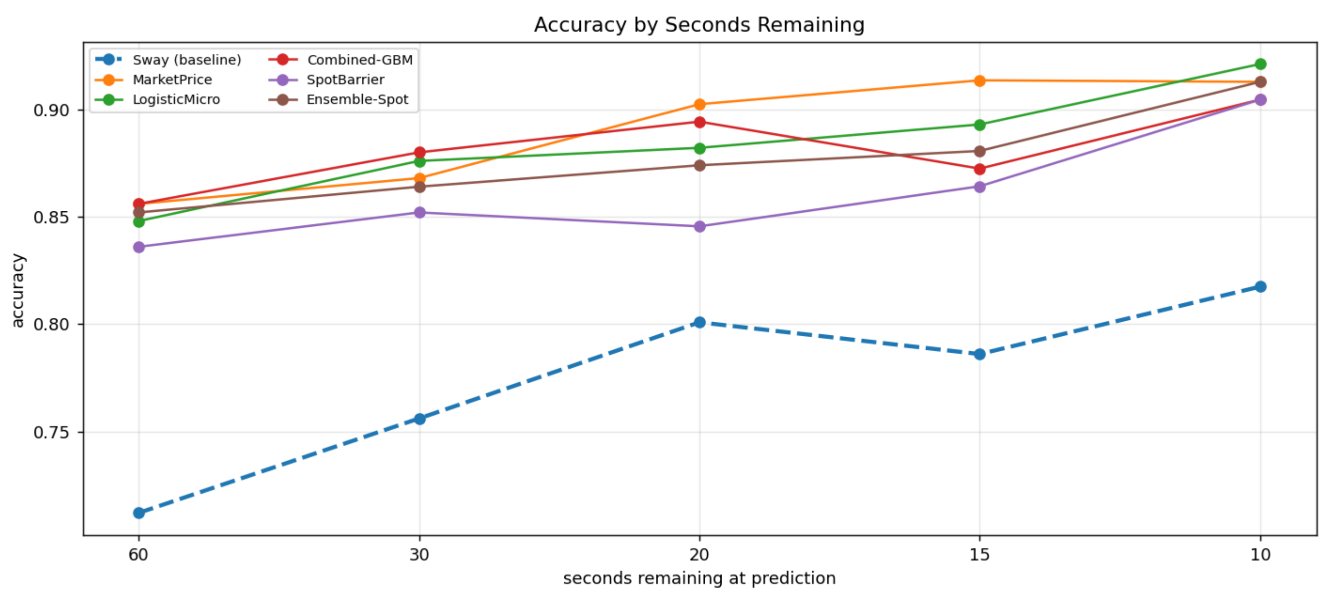
| Strategy | Accuracy | Brier | LogLoss | Bets | ROI | P&L $ | WinRate | MaxDD$ |
|---|---|---|---|---|---|---|---|---|
| **MarketPrice** | **89.0%** | **0.0800** | **0.2547** | **0** | **+0.0%** | **+0.0** | **0.0%** | **0.0** |
| XGB-Calibrated | 88.9% | 0.0868 | 0.2867 | 601 | -8.4% | -50.4 | 47.6% | 132.6 |
| RF-Rich | 88.9% | 0.0803 | 0.2569 | 476 | -2.1% | -10.1 | 71.0% | 44.4 |
| Edge-Ridge | 88.8% | 0.0822 | 0.3140 | 620 | -29.0% | -179.5 | 49.0% | 182.2 |
| Momentum | 88.5% | 0.0828 | 0.3403 | 370 | -8.9% | -33.1 | 63.8% | 46.7 |
| LogisticMicro | 88.4% | 0.0849 | 0.2768 | 548 | +3.3% | +17.9 | 70.3% | 33.7 |
| Combined-GBM | 88.1% | 0.0900 | 0.2914 | 647 | +13.1% | +84.9 | 59.8% | 57.7 |
| Edge-GBM | 88.0% | 0.0870 | 0.3211 | 540 | -13.9% | -75.0 | 54.6% | 84.9 |
| GBM-Rich | 87.9% | 0.0895 | 0.2928 | 603 | -7.8% | -47.1 | 61.5% | 77.4 |
| XGB-Rich | 87.9% | 0.0919 | 0.3069 | 613 | -0.9% | -5.5 | 69.2% | 49.9 |
| LGBM-Rich | 87.9% | 0.0977 | 0.3400 | 632 | -2.0% | -12.5 | 70.4% | 36.2 |
| Ensemble-Spot | 87.6% | 0.0848 | 0.2658 | 562 | +20.5% | +115.5 | 62.1% | 48.3 |
| Combined-Logistic | 87.6% | 0.0870 | 0.2877 | 588 | +13.5% | +79.5 | 68.7% | 25.1 |
| Logistic-Calib | 87.5% | 0.0862 | 0.3023 | 538 | -12.4% | -66.8 | 50.7% | 80.6 |
| Logistic-PerSlot | 87.2% | 0.0925 | 0.3193 | 593 | -1.4% | -8.4 | 67.5% | 33.2 |
| SpotBarrier | 86.0% | 0.1178 | 0.8329 | 652 | +20.6% | +134.3 | 64.0% | 49.0 |
| SpotML-Logistic | 84.5% | 0.1099 | 0.3440 | 888 | -21.7% | -192.6 | 23.2% | 228.9 |
| Sway (baseline) | 77.4% | 0.1534 | 0.5624 | 890 | -27.3% | -242.9 | 28.2% | 288.5 |
| SpotML-GBM | 76.7% | 0.1472 | 0.4411 | 747 | -4.6% | -34.2 | 35.3% | 119.7 |

*Sway baseline highlighted in red; best strategy in bold. Trading simulation bets $1 whenever a model's probability diverges from the live market price by > 3% (positive expected value), including Polymarket fees.*

# Accuracy & Calibration

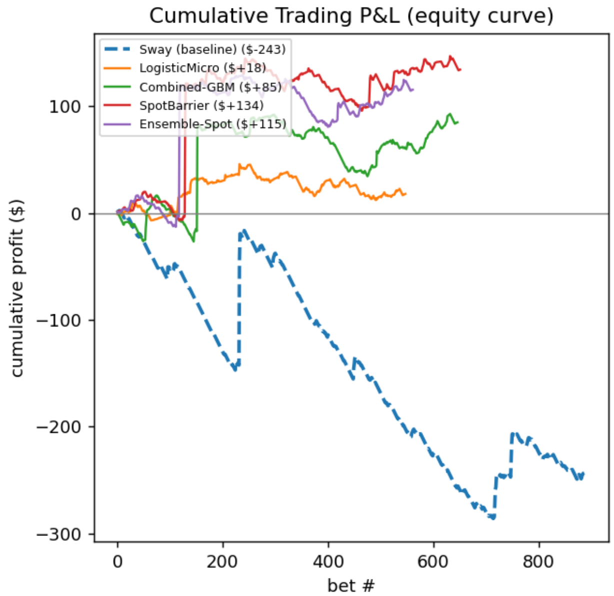


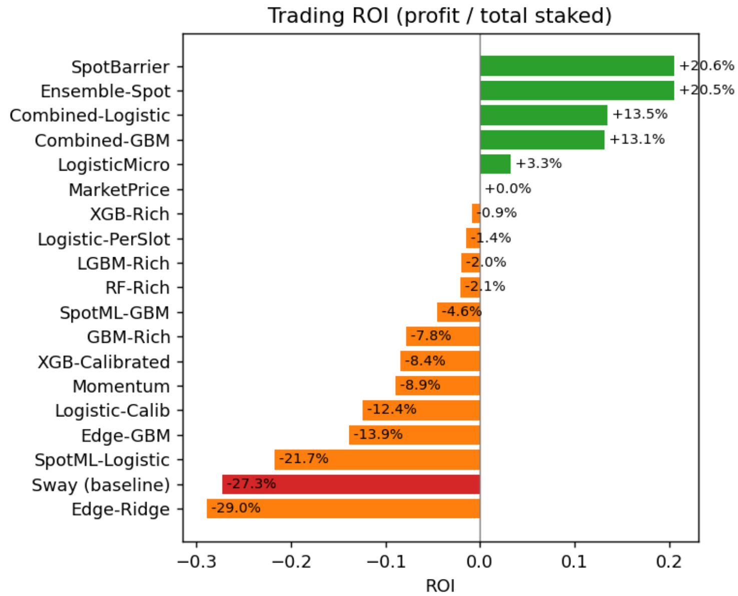



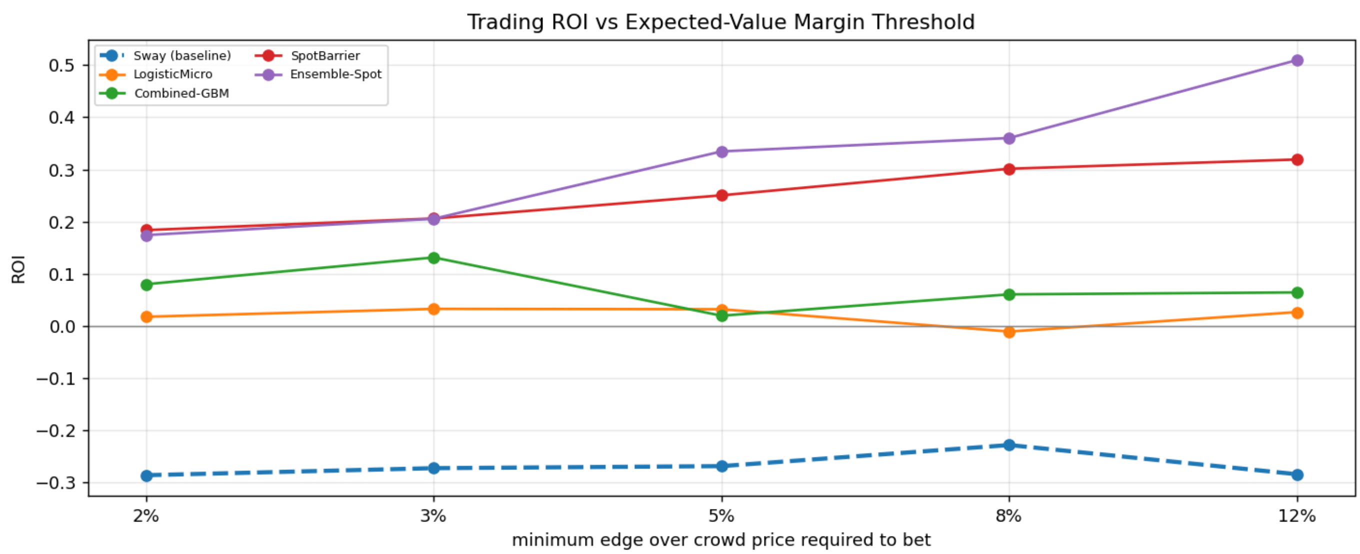



## Trading Performance

The sway model ignores the absolute market price, so its statistical edge does not always convert into trading profit. The equity curves below show realised P&L when each model only bets on positive-EV divergences from the crowd price.



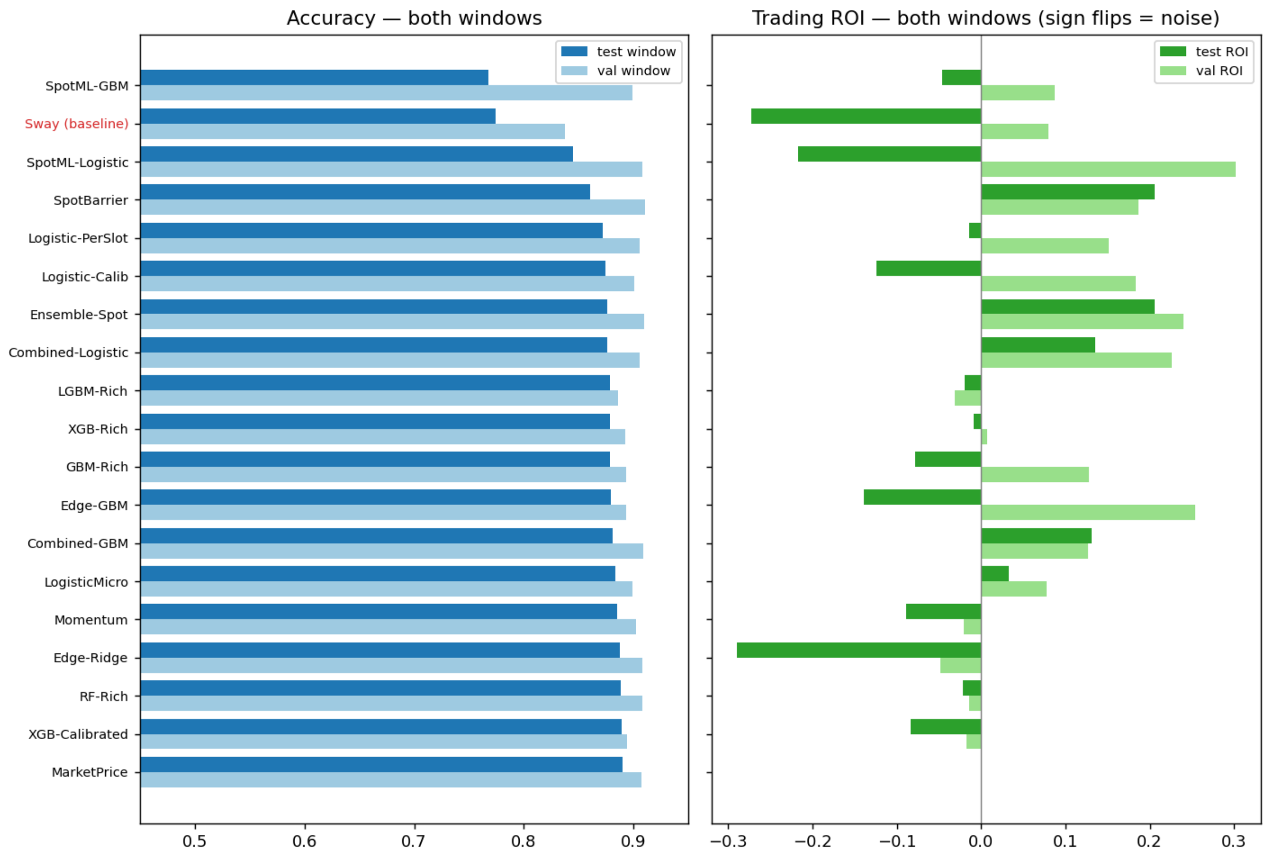



## Cross-Window Robustness

To separate genuine edge from luck, every strategy is re-evaluated on a second, fully independent and time-disjoint window of **300** older markets, in a different regime (45.7% UP vs 53.2% in the recent test window).

**Two clear conclusions:** (1) Statistical quality is robust — the Sway baseline is the least accurate model in *both* windows by a wide margin. (2) Single-window trading ROI is largely noise: most prediction-market-only strategies flip sign between windows (e.g. Edge-GBM from -14% to +25%). (3) The spot-driven strategies are different: they are profitable in *both* windows. Strategies with **positive ROI on both** windows (ranked by worst-case ROI): **Ensemble-Spot, SpotBarrier, Combined-Logistic, Combined-GBM, LogisticMicro**. The analytic SpotBarrier model leads — a repeatable, sizeable trading edge.



## Methodology

All strategies use only trade data available up to the prediction time (no look-ahead). Predictions are made at 60, 30, 20, 15 and 10 seconds remaining. The Sway baseline is a faithful reproduction of the repo's per-slot GradientBoosting model on the 29 channel-sway features. Prediction-market candidates add the absolute price level, VWAP, multi-horizon momentum/volatility and order-flow imbalance. **Spot-driven candidates** additionally read the underlying BTC price (Binance 1s klines): the SpotBarrier model computes an analytic first-passage probability $P(\text{close} > \text{open}) = \Phi(\text{lead} / (\sigma\sqrt{t\_\text{remaining}}))$ from the live spot lead and realised volatility; the Combined models feed spot + market features into a classifier. Metrics: directional accuracy, Brier score and log-loss (probability quality), plus a fee-aware Polymarket P&L simulation that only bets on positive-EV divergences from the crowd price.